

Time Series: Some Basic Concepts

- (A) Introduction: Time Series
- (B) Classical Linear Model Assumptions in Time Series Analysis
- (C) Static Model
- (D) Finite Distributed Lags Model

(A) Introduction: Time Series

- Time series data set comes with a temporal ordering. One is immediately preceded by another. For example, the inflation rate for 1991 October immediately precedes the inflation rate for 1991 November, the inflation rate for 1991 November precedes the inflation rate for 1991 December, etc.

1991 M10	2.4
1991M11	2.5
1991 M12	2.9
1992M1	2.8
⋮	⋮
1998M6	0.2
1998M7	0.5
1998M8	-1
⋮	⋮
2008M7	6.5
2008M8	6.4



- The order (and temporal relationships) is important as it tells us a great information. So, when modeling relationships between variables, the nature of the data has an important bearing on the appropriate choice of an econometric model. Therefore, it is important to distinguish between cross-sectional data panel data and time series data.
- For time series data, there is a natural ordering; the observations are ordered according to time. More specifically, the time series data is a sequence of observations taken over time: y_t where $t = 1, 2, \dots, T$. That is $y_1, y_2, \dots, y_T$.

- Although time is continuous, the interval between observations is discrete. Usually, the time interval of the observations is equally spaced. The interval can be yearly, quarterly, monthly, daily, hourly, etc.

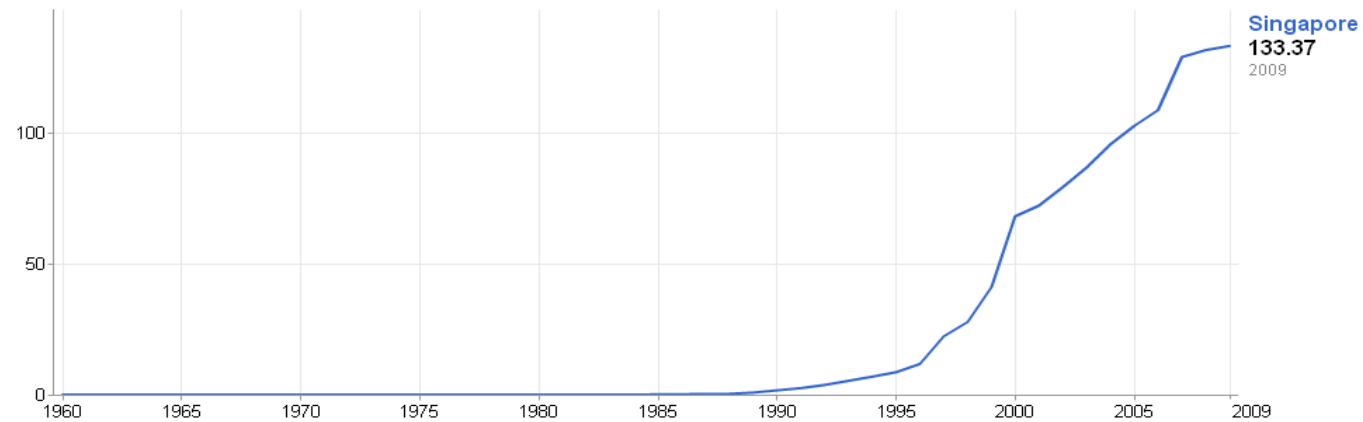
Example:

- (a) GDP: quarterly or yearly data
- (b) inflation rate or unemployment rate: monthly, quarterly, yearly data
- (c) stock index or exchange rate: daily, weekly, monthly, quarterly, yearly data
- (d) company earnings, dividends, etc.: quarterly, yearly
- (e) temperature: hourly data

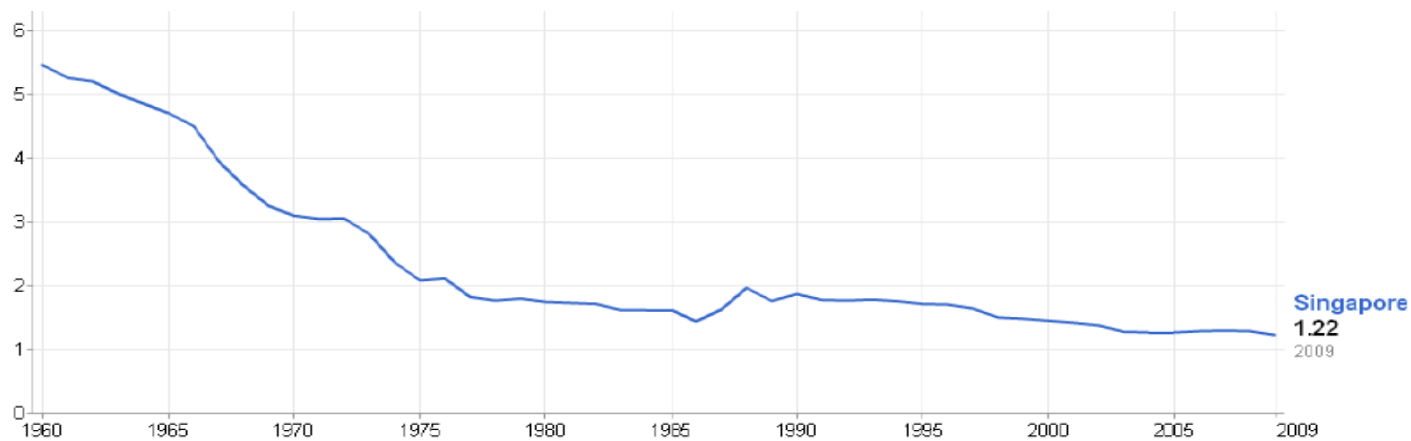
- Discussion:
 - (a) The frequency of the data needs to be consistent for all variables when we are running a regression model.
 - (b) To sum up 4 quarter data for any given years = yearly data. Similarly, if we have daily data, we can convert the frequency of the time series. For instance, we have daily exchange rate data, and we can convert the frequency to month.
 - (c) Suppose we have quarterly data. We can also change the frequency of the time series to month. For example, we can use cubic spline interpolation.

Time series graphs from the internet

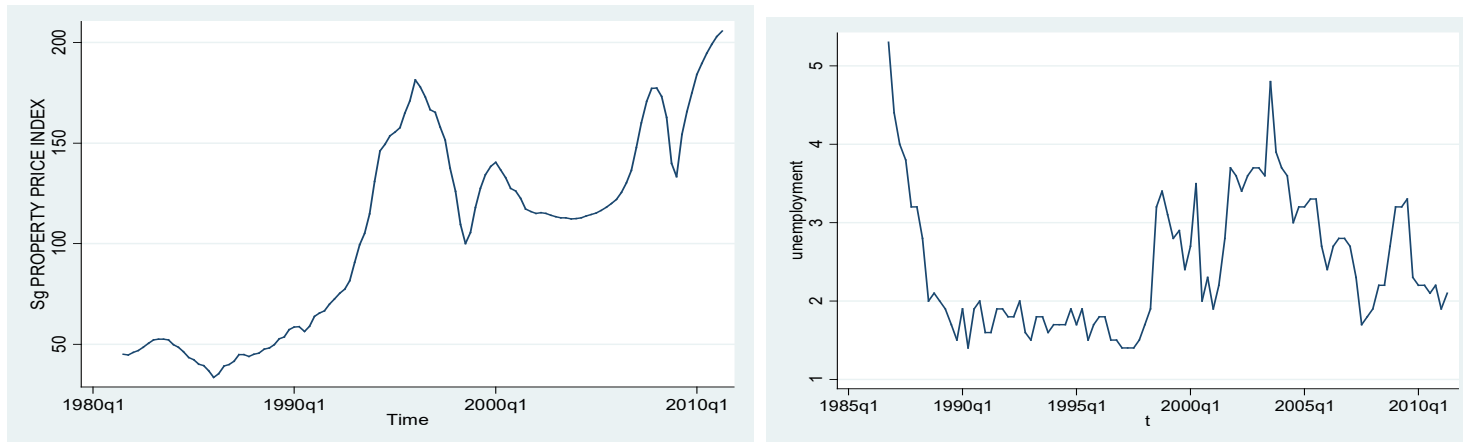
[1] Mobile subscriptions (per 100 people)



[2] Singapore fertility rate



[3] Singapore Private Housing Price Index and Unemployment Rate



[LEFT] Around 1996, the index reached a new high. In 1997 July, the Asian financial crisis hit Singapore, the index started to drop until 1998 third quarter. In the second quarter of 2010, the index eventually went back to the pre-crisis level.

[RIGHT] Singapore experienced a recession in year 1985.

- Rise in labor costs & shortage of skilled workers.
- A strong fierce competition from Middle Eastern companies on oil refining & industry started in year 1984
- Regional Recessions

2003 October, the Singapore unemployment rate hit a new high compared to its past 17 years.

- The sequence of a time series random variable (e.g., mobile subscriptions, fertility rate, housing price index, unemployment rate) indexed by time is called stochastic process or a time series process because the future is uncertain. We don't know the value associated with the random variable if it has not happened yet.
- When collecting a time series data set, we obtain one possible outcome (or realization) of the stochastic process. We can only *have* one realization because we don't have a time machine to back in time and start the process over again
- For the cross-sectional data, we can recollect the data from the population.
- Unlike the cross-sectional data, for the time series data, unless we can find another parallel world, we cannot have another realization.

- An intrinsic feature of a time series, e.g., fertility rate, housing price, unemployment rate, is that the adjacent observations are often dependent.
- (Applied) time series analysis is concerned with techniques for the analysis of this dependence. Speaking differently, we are concerned with the dynamic consequences of events over time.

Some differences between cross-section and time-series data structure

	cross-sectional units	time series units
Time	Time dimension is not involved.	Time dimension is important
Subscript	$i = 1, 2, \dots, n$ i indexes different persons, workers, firms, cities, etc.	$t = 1, 2, \dots, n$ t indexes different time periods.
Correlations	no correlation between observations from a random sample. So, we can rearrange the order of the observations, and it does not destroy the information	Observations of a variable are usually auto-correlated. If we rearrange the order of the observations, the structure is destroyed.
Outcome of the samples	random sample	random outcome
	a collected sample	a realization

(B) Classical Linear Model Assumptions in Time Series Analysis

- In general, we can use the ordinary least squares method to analyze the time series data, because the OLS has a number of desirable properties and also the hypothesis tests regarding the estimates could be validly conducted.

ASSUMPTION TS.1 (LINEAR IN PARAMETERS)

The stochastic process $\{(x_{t1}, x_{t2}, \dots, x_{tk}, y_t): t = 1, 2, \dots, n\}$ follows the linear model

$$y_t = \beta_0 + \beta_1 x_{t1} + \dots + \beta_k x_{tk} + u_t, \quad (10.8)$$

where $\{u_t: t = 1, 2, \dots, n\}$ is the sequence of errors or disturbances. Here, n is the number of observations (time periods).

- Assumption TS. 1 is essentially same as the multiple linear regressions assumption 1 for the cross sectional models. It is very easy to check the assumption TS1.

Example: (a), (b), and (c) satisfy Assumption TS. 1

$$(a) \ln(y_t) = \beta_0 + \beta_1 \ln(x_{t1}) + \beta_2 x_{t2} + u_t$$

$$(b) \ln(y_t) = \beta_0 + \beta_1 \ln(x_t) + \beta_2 \ln(y_{t-1}) + u_t$$

$$(c) y_t = \beta_0 + \beta_1 \ln(x_{t1}) + \beta_2 \ln(x_{t2}) + u_t$$

- **Assumption TS. 2: Zero Conditional Mean:** For each t , the expected value of the error u_t , given the explanatory variables for all time periods is zero. Mathematically, $E(u_t|\mathbf{X}) = 0, t = 1, 2, \dots, n$, or

$$E(u_t|x_1, x_2, \dots, x_n) = 0, t = 1, 2, \dots, n$$

- This is the crucial assumption in time series analysis. Assumption TS. 2 implies that the (random) error, u_t at time t , is uncorrelated with all explanatory variables in every time period. Clearly, Assumption TS. 2 is a very strong assumption. Often this is an unrealistic assumption. It requires that u_t not only is uncorrelated with current explanatory variable $\mathbf{x}$ but also uncorrelated with the past and future values $\mathbf{x}$. [Strict Exogeneity].

For example: Consider $y_t = \beta_0 + \beta_1 x_{t1} + \beta_2 x_{t2} + u_t$, where $t = 1, 2, \dots, n$

$$t = 1 \quad : \quad y_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + u_1,$$

$$t = 2 \quad : \quad y_2 = \beta_0 + \beta_1 x_{21} + \beta_2 x_{22} + u_2,$$

$$t = 3 \quad : \quad y_3 = \beta_0 + \beta_1 x_{31} + \beta_2 x_{32} + u_3,$$

$$\vdots$$

$$t = 10 \quad : \quad y_{10} = \beta_0 + \beta_1 x_{10,1} + \beta_2 x_{10,2} + u_{10},$$

$$\vdots$$
$$\vdots$$
$$\vdots$$

$$t = (n - 2): y_{(n-2)} = \beta_0 + \beta_1 x_{(n-2),1} + \beta_2 x_{(n-2),2} + u_{(n-2)},$$

$$t = (n - 1): y_{(n-1)} = \beta_0 + \beta_1 x_{(n-1),1} + \beta_2 x_{(n-1),2} + u_{(n-1)},$$

$$t = n \quad : \quad y_n = \beta_0 + \beta_1 x_{n,1} + \beta_2 x_{n,2} + u_n.$$

(a) The strict exogeneity assumption requires the error term for example u_2 not to be correlated with the explanatory variables x_1 and x_2 at every period.

(b) The matrix expression is $y_t = \mathbf{x}_t' \boldsymbol{\beta} + u_t$, where $t = 1, 2, \dots, n$, $\mathbf{x}$ is a 3×1 matrix and $\boldsymbol{\beta}$ is also a 3×1 matrix. The strict exogeneity means

$$E(u_t | \mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n) = 0, t = 1, 2, \dots, n$$

Question: Can you give one example that violates the strict exogeneity assumption?

- Recap: Consider a cross-sectional model. We only need the error u_i not to be correlated with x_i so that the estimators are unbiased. For example,

$$\ln(wage_i) = \beta_0 + \beta_1 educ_i + \beta_2 exper_i + u_i, \text{ where } i = 1, 2, \dots, n$$

- A weaker version of Assumption TS. 2 is $E(u_t | \mathbf{x}_t) = 0$. Here u_t and the explanatory variables $\mathbf{x}_t$ are contemporaneously uncorrelated, i.e.,

$$cov(x_{t1}, u_t) = 0, cov(x_{t2}, u_t) = 0, \dots, cov(x_{tk}, u_t) = 0.$$

If the unobservable factors at the time t , u_t , are uncorrelated with all the explanatory variables $\mathbf{x}_t$ at the time t , we say the contemporaneous exogeneity assumption holds.

- **Assumption TS. 3: No Perfect Collinearity:** In the sample (and therefore in the underlying time series process), none of the explanatory variables is constant or a perfect linear combination of the other explanatory variables.

If Assumption T3. 3 is violated, how do we fix it?

- (a) We should drop one of the perfect collinear explanatory variables from the model.
- (b) In general, the problem arises because there are too many dummy variables created and added to the regression model. Therefore, we can drop one dummy variable.

- **Theorem: Unbiasedness of OLS** (for time series data): Under Assumptions TS. 1, 2, and 3 the OLS estimators are unbiased, $E(\hat{\beta}_j | \mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n) = \beta_j$, where $j = 1, 2, \dots, k$

- **Assumption TS. 4: Homoskedasticity:** The variance of u_t conditional on $\mathbf{X}$ is constant of all t , i.e.,

$$\text{Var}(u_t|\mathbf{X}) = \text{Var}(u_t) = \sigma^2, \quad t = 1, 2, \dots, n$$

- Heteroskedasticity can also occur in time series regression models; its presence, while not causing bias nor inconsistency in the point estimates, has the usual effect of invalidating the standard errors, t -statistics, and F -statistics, just as in the cross-sectional case.
- Assumption TS. 4 implies $\text{Var}(u_t|\mathbf{X})$ does not depend on $\mathbf{X}$. If the homoskedasticity assumption is violated, the model contains heteroskedastic errors.

Example:

$$\text{var}(u|x) = \sigma^2 \cdot f(x)$$

- **Assumption TS. 5: No Serial Correlation**

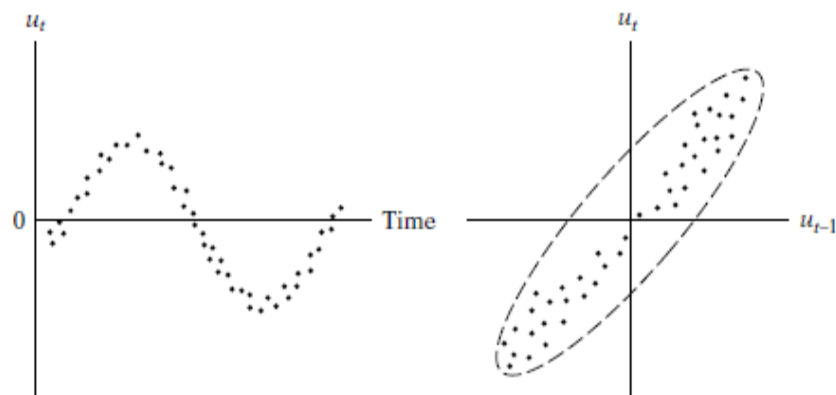
Conditional on $\mathbf{X}$, the errors from any two different time periods are uncorrelated, i.e.,

$$\text{Corr}(u_h, u_s | \mathbf{X}) = 0 \quad \forall h \neq s.$$

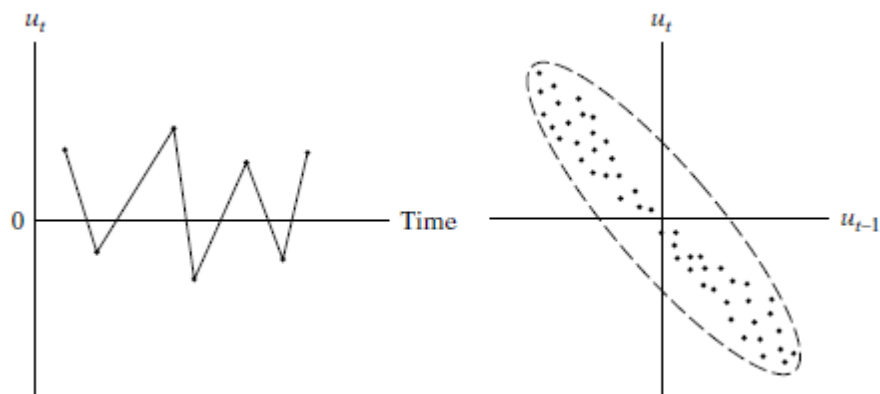
- If the no serial correlation assumption is violated, then the errors in the regression models suffer from the serial correlation (or autocorrelation) problem. For example, when u_t and u_{t-1} are correlated, they are autocorrelated
- In the cross-sectional analysis, the individual (or firm/ country) i is not related to the individual (or firm/ country) j from a random sample. Therefore, this assumption is not needed.
- However, $\text{Cov}(u_i, u_j) \neq 0$ often exists in the time series data. The unobserved effect will generally be correlated.

- Compared to heteroskedasticity, the serial correlation problem creates a bigger impact on the standard errors of the estimators

(a) The diagrams below show that there is a positive (auto)correlation between u_t and u_{t-1} .



(b) The diagrams below shows that there is a negative (auto)correlation between u_t and u_{t-1} .



- **Gauss-Markov Theorem:** Under Assumptions TS. 1 to TS. 5, the OLS estimators are the best linear unbiased estimators (BLUE) conditional on $\mathbf{X}$.
- **Assumption TS. 6: Normality:** The errors u are independent of $\mathbf{X}$ and are independently and identically distributed as normal $(0, \sigma^2)$.
- **Theorem (Normal Sampling Distribution):** Under Assumptions TS1. Through TS. 6 (known as the classical linear model assumptions for time series), the OLS estimators are normally distributed, conditional on $\mathbf{X}$.
- **Discussion**
 - (a) Assumption TS. 6 implies Assumptions TS. 2, 4, and 5. Assumption TS. 6 is stronger because of the independence and the normality assumption.
 - (b) The usual construction of confidence intervals and the t and F tests (covered in HE2003/HE2005) applies directly to time series regression.

(c) However, the classical linear model assumptions for time series data are *much more restrictive* than those for cross-sectional data. Especially, the strict exogeneity assumption (TS. 2) and no serial correlation (TS. 5) are often violated.

(d) If either one of the assumptions, heteroskedasticity and autocorrelation, is violated, the usual OLS estimators do not have the minimum variance among all other competing linear unbiased estimators. In addition, the usual statistical tests, e.g., the t and F tests, are invalid.

(C) Static Model

- Consider two time series variables, say y and x_1 , where y_t and x_{t1} are dated contemporaneously. A static model relating y to x_1 is

$$y_t = \beta_0 + \beta_1 x_{t1} + u_t, \text{ where } t = 1, 2, \dots, n$$

- This is the simplest time series model, closest to the cross-sectional models. The regression models the contemporaneous relationship between y and x_1 , and a change in x_1 have an immediate effect on y .

Example:

(a) $y_t = \beta_0 + \beta_1 x_{t1} + \beta_2 x_{t2} + u_t$, where $t = 1, 2, \dots, n$

(b) $y_t = \beta_0 + \beta_1 x_{t1} + \beta_2 x_{t2} + \dots + \beta_k x_{tk} + u_t$, where $t = 1, 2, \dots, n$

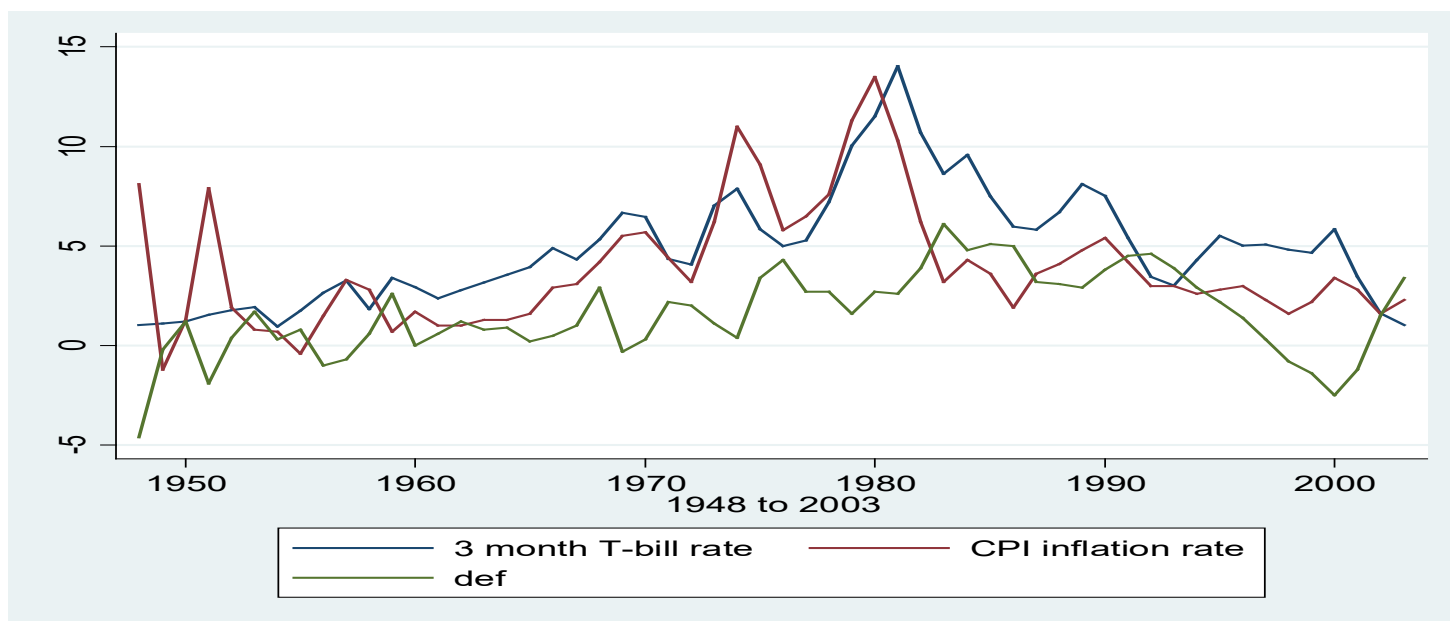
(c) $y_t = \beta_0 + \beta_1 x_{t1} + \beta_2 t + u_t$, where $t = 1, 2, \dots, n$

(d) $stock_price_t = \beta_0 + \beta_1 interest_rate_t + u_t$, where $t = 1, 2, \dots, n$

Example: Effects of inflation and deficit on interest rates: The INTDEF data is obtained from the 2004 Economic Report of the President and span the years 1948 through 2003. The variable $i3$ is the three-month T-bill rate, inf is the annual inflation rate based on the consumer price index (CPI), and def is the federal budget deficit as a percentage of GDP. Consider a static regression model as follows

$$i3_t = \beta_0 + \beta_1 inf_t + \beta_2 def_t + u_t, \text{ where } t = 1948, 1949, \dots, 2003$$

>tsline i3 inf def



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. reg i3 inf def
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Source	SS	df	MS	Number of obs =	56
Model	272.420338	2	136.210169	F(2, 53) =	40.09
Residual	180.054275	53	3.39725047	Prob > F =	0.0000
				R-squared =	0.6021
				Adj R-squared =	0.5871
Total	452.474612	55	8.22681113	Root MSE =	1.8432

i3	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
inf	.6058659	.0821348	7.38	0.000	.4411243 .7706074
def	.5130579	.1183841	4.33	0.000	.2756095 .7505062
_cons	1.733266	.431967	4.01	0.000	.8668497 2.599682

These estimates show that increases in inflation and the relative size of the deficit work together to increase short-term interest rates, both of which are expected from basic economics. For example, a ceteris paribus one percentage point increase in the inflation rate increases *i3* by .605 points. Both *inf* and *def* are statistically significant, assuming, of course, that the classical linear model assumptions hold.

- **Discussion:**

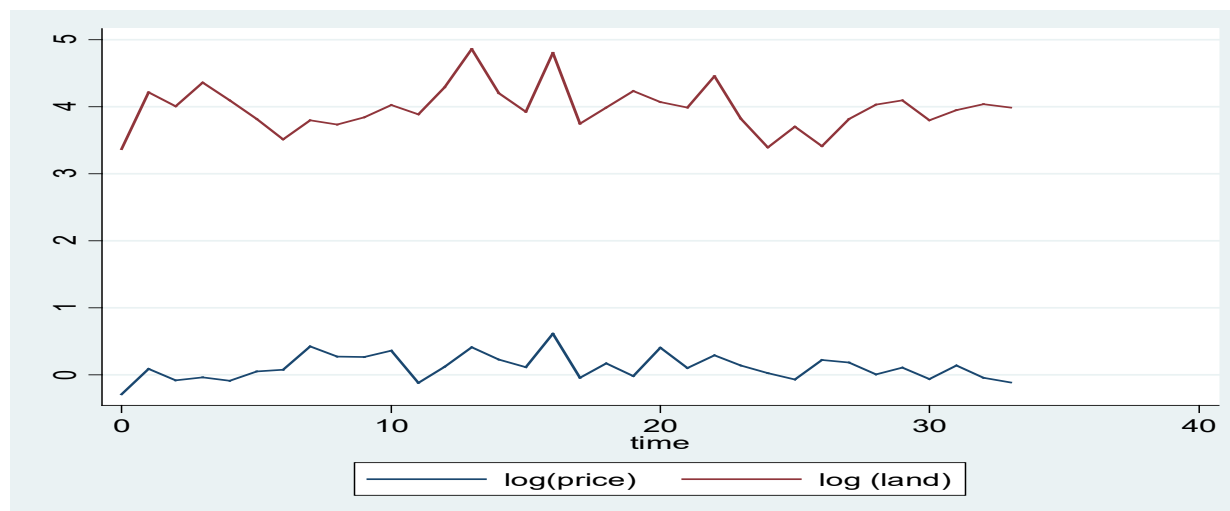
In practice, we should always test the stationarity of the time series data before running a regression model. If the time series data is not stationary, then we cannot run the regression model using the raw data directly. For this empirical example, all variables are stationary even though we do not tests for the stationarity here

Example: When price of sugarcane is high, farmers plant more than when its price is low. We can estimate the relationship between size of the planted area and price.

$$\ln(\text{land}_t) = \beta_0 + \beta_1 \ln(\text{price}_t) + u_t$$

Information on β_1 is useful for government planning. Because it is important to know whether existing sugar processing mills are likely to be able to handle predicted output, whether a pricing policy linking production, processing, and consumption is desirable.

>tsset time **>tsline** Lprice Lland



. reg Lland Lprice

Source	SS	df	MS	Number of obs = 34		
Model	.741225795	1	.741225795	F(1, 32) = 7.82		
Residual	3.03157079	32	.094736587	Prob > F = 0.0087		
				R-squared = 0.1965		
				Adj R-squared = 0.1714		
Total	3.77279658	33	.114327169	Root MSE = .30779		

Lland	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
Lprice	.7761187	.2774673	2.80	0.009	.2109364	1.341301
_cons	3.893256	.0613451	63.46	0.000	3.7683	4.018212

How do we interpret the estimated coefficient of Lprice?

Notes:

- (a) We haven't checked the stationarity of the time series data. We must make sure the series are stationary. Otherwise the regression analysis is meaningless.
- (b) We assume the TS. 4 and TS. 5 assumptions hold for now. But, if the assumptions are not satisfied, then the standard errors are wrong. And, we need to adjust the wrong standard errors. (Recall the case in cross-sectional models.)

(D) Finite Distributed Lags Model

- In a finite distributed lag (FDL) model, we allow one or more variables to affect y with a lag. For example, the model below is a FDL of order 2.

$$y_t = \alpha_0 + \delta_0 x_t + \delta_1 x_{t-1} + \delta_2 x_{t-2} + u_t, \text{ where } t = 3, 2, \dots, T$$

- The slopes δ_0 , δ_1 , and δ_2 indicate the impact of a unit change of x on y . δ_0 is called the impact multiplier.
- For a FDL of order p, the model can be expressed as follows.

$$\begin{aligned} y_t &= \alpha_0 + \delta_0 x_t + \delta_1 x_{t-1} + \delta_2 x_{t-2} + \dots + \delta_p x_{t-p} + u_t \\ &= \alpha_0 + \sum_{s=0}^p \delta_s x_{t-s} + u_t, \end{aligned}$$

One difficulty common to all distributed-lag models is find an optimal number of the lag length. What is the number for p ?

Example:

What is the impact of x on y when at the time t , x_t increases 1 unit temporary from “c” to “c+1” and then goes back to a constant “c” after time t ? For the sake of simplicity, we assume that the error terms are zero.

$$\text{At time } (t - 1): y_{(t-1)} = \alpha_0 + \delta_0 x_{t-1} + \delta_1 x_{t-2} + \delta_2 x_{t-3} = \alpha_0 + \delta_0 c + \delta_1 c + \delta_2 c$$

$$\text{At time } t : y_t = \alpha_0 + \delta_0 x_t + \delta_1 x_{t-1} + \delta_2 x_{t-2} = \alpha_0 + \delta_0 (c + 1) + \delta_1 c + \delta_2 c$$

$$\text{At time } (t + 1): y_{(t+1)} = \alpha_0 + \delta_0 x_{t+1} + \delta_1 x_t + \delta_2 x_{t-2} = \alpha_0 + \delta_0 c + \delta_1 (c + 1) + \delta_2 c$$

$$\text{At time } (t + 2): y_{(t+2)} = \alpha_0 + \delta_0 x_{t+2} + \delta_1 x_{t+1} + \delta_2 x_t = \alpha_0 + \delta_0 c + \delta_1 c + \delta_2 (c + 1)$$

$$\text{At time } (t + 3): y_{(t+3)} = \alpha_0 + \delta_0 x_{t+3} + \delta_1 x_{t+2} + \delta_2 x_{t+1} = \alpha_0 + \delta_0 c + \delta_1 c + \delta_2 c$$

- **Discussion**

- (a) A temporary change on x at time t only has an impact multiplier δ_0 on the y variable.
- (b) After the time t , say at the time $(t + 1)$, the $y_{(t+1)}$ is affected by the change of the x at time t . $y_{(t+1)}$ increases by δ_1 unit on average.
- (c) At the time $(t + 2)$, the $y_{(t+2)}$ is still affected by the change of x two periods ago. $y_{(t+2)}$ increases by δ_2 unit on average
- (d) At the time $(t + 3)$, change on x at time t doesn't affect the $y_{(t+3)}$ anymore.
- (e) The cumulative effect, is $\delta_0 + \delta_1 + \delta_2$

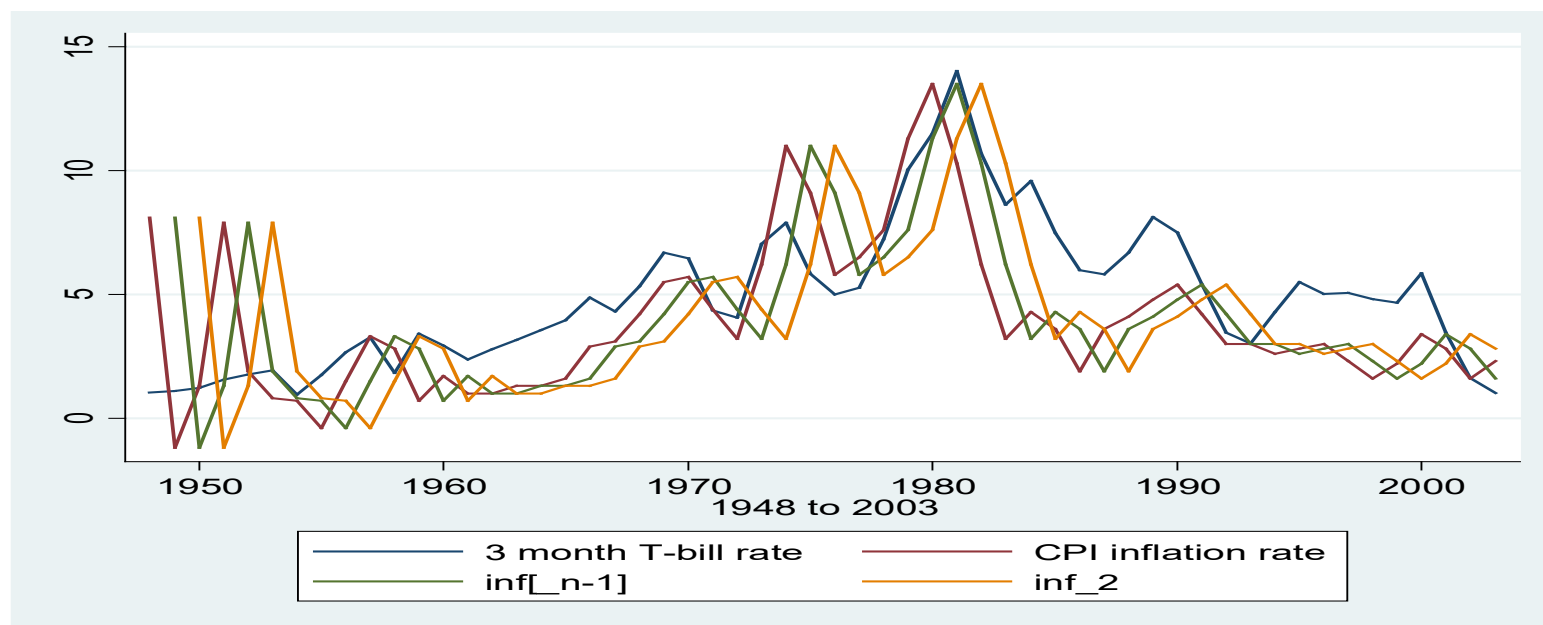
Example: In an equation for annual data from 1948 to 2003, suppose that

$$interest_t = \alpha_0 + \delta_0 inf_t + \delta_1 inf_{t-1} + \delta_2 inf_{t-2} + u_t,$$

where $t = 1948, 1949, \dots, 2003$

interest is an interest rate and *inf* is the inflation rate. What are the impact propensity and long-run propensity?

>**tsset** time; >**tsline** interest inf L1.inf L2.inf



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. reg i3 inf L1.inf L2.inf
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Source	SS	df	MS	Number of obs = 54		
Model	260.918294	3	86.9727646	F(3, 50) = 27.01		
Residual	160.999489	50	3.21998979	Prob > F = 0.0000		
				R-squared = 0.6184		
				Adj R-squared = 0.5955		
Total	421.917783	53	7.96071289	Root MSE = 1.7944		

i3	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
inf	.4564295	.1297251	3.52	0.001	.1958689	.7169901
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L1.	.1809089	.1551053	1.17	0.249	-.1306293	.492447
L2.	.2090695	.1110873	1.88	0.066	-.0140559	.4321948
_cons	1.74949	.4501129	3.89	0.000	.8454119	2.653569

Note: L1.X represent the x variable in one lag, i.e., x_{t-1} . Similarly, Lj.X represent the x variable in j lag, i.e., x_{t-j} .

The impact propensity is 0.456;. The long-run propensity (or long-run multiplier) is just the sum of the estimated coefficients associated with the inflation and its lags,

$$LRP = \hat{\delta}_0 + \hat{\delta}_1 + \hat{\delta}_2 = \underline{0.845}$$